

STUDIO ALICE

An in-depth look at Alice, the

codename for Ubisoft's screen-writing room. A behind the scenes glimpse of the creative process behind Ubisoft titles such as FarCry, Assassin's Creed, etc. <http://dgit.in/13inELi>

PS4: THE LAST CONSOLE?

The PlayStation 4 launch event was a bummer as the actual PS4 was not unveiled. This Wired article goes on to debate why according to them, the PS4 is probably the last of the console games that will ever be released, with more stress on all-in-one entertainment in the future iterations. <http://dgit.in/XBxAZv>

UP CLOSE WITH THE NEURONS

This gallery on Wired may look like microscopic

photographs of the neurons, but are actually imaginative, yet knowledgeable paintings by neuroscientist Greg Dunn <http://dgit.in/VIOHuS>

NEW GESTURE INTERFACE

What we saw in this video made us think that it was

a spoof, but the MYO - Wearable gesture control from Thalmic Labs is actually a \$149 armband for gesture control. <http://dgit.in/W80BQ0>

larger buffers to allow for higher burst bandwidth and more aggressive power savings.

TEGRA 4'S CPU

Nvidia will launch two flavors of Tegra 4 this year. The full-size version will combine a quad-core Cortex A-15 with a 72-core GPU, while Tegra 4i will use a quad-core Cortex-



A9 R4 processor and a 60-core graphics processor. The R4 variant of the classic A9 architecture is a custom chip Nvidia has built to increase the performance of the older CPU. Tegra 4i will have a 32K L1 data cache (standard A9s only had 16K), an L1 data prefetcher, dedicated prefetch buffers, a larger translation lookaside buffer, and better branch prediction performance.

Tegra 4 implements a 32-instruction loop buffer that allows it to shut off some of the Cortex-A15's most expensive logic blocks as far as power consumption is concerned. According to the whitepaper, fetches, decodes, and branch prediction consume roughly 40% of the CPUs power budget. Power gating those elements when not in use significantly improves power consumption.

NV has apparently changed the name of the Companion Core once again — it now refers to the chip as a Battery Saver Core.

Companion Core is still an infinitely better option, and both of these quad-cores contain one. We've covered most of the features of the Cortex-A15 in previous articles, but Nvidia makes much of how its CPU design is superior to Krait. That's no accident — Qualcomm is going to be Nvidia's toughest competition for business this time out. Nvidia is hoping its Icera softmodem can win space away from Qualcomm's 28nm integrated LTE, despite the fact that Tegra 4's CPU + modem combination is fairly large for a mobile chip, while Tegra 4i is based on the much older A9.

The A9 R4 is actually quite an interesting chip, though we may not see much market for it in the United States. As we've previously discussed, the Cortex-A15 isn't a great smartphone processor; devices like the Nexus 4 struggle to keep the chip from drawing too much power and overheating. Refining the Cortex-A9's design further by incorporating certain aspects of the A15 may be a potent way to improve mid-range smartphone performance without sacrificing battery life or capabilities. At the very least, the

new quad-core A9 R4 should outperform Tegra 3's CPU — and that chip has proven to be quite popular in tablets.

PUTTING IT ALL TOGETHER

Some of the leaked benchmark results show Tegra 4 destroying modern competition, though I'll wait for shipping



hardware to make a final determination. It seems clear, however, that T4 is a significant step forward. It's the first design that puts Nvidia's Icera modem front and center and it should help re-establish Tegra as a smartphone brand. Tegra 3's sales were largely driven by tablets; smartphone shipments were relatively small. Tegra 4i's move to a 28nm process will lower power consumption enough to give the chip smartphone options, and we expect it to do well in that space.

The games Nvidia is showing off as demonstrations of what's possible between Tegra 3 and Tegra 4 illustrate significant updates between the two graphics cards. Lighting and shadowing are considerably improved, though they still conform to DX9 standards.

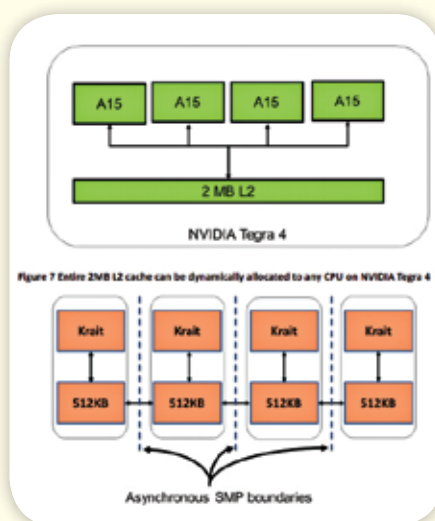


Figure 7 Entire 2MB L2 cache can be dynamically allocated to any CPU on NVIDIA Tegra 4